INSA INSTITUT NATIONAL DES SCIENCES APPLIQUÉES LYON

SemiStructQuest: unveiling NoSQL database concepts through gamified learning platform analyses

Nelly BARRET and Mélina VERGER

INSA Lyon and LIRIS lab, France

Workshop DataEd'26 – March 24th 2026

CONTEXT: NOSQL DATABASES

- NoSQL databases are ubiquitous
- Part of most STEM diplomas

CONTEXT: NOSQL DATABASES

- NoSQL databases are ubiquitous
- Part of most STEM diplomas

- Traditional learning is predominant for NoSQL courses
  - Lectures, written exercises, hand-on projects
- Few guidelines / little perspective

CONTEXT: NOSQL DATABASES

- NoSQL databases are ubiquitous
- Part of most STEM diplomas

- Traditional learning is predominant for NoSQL courses
  - Lectures, written exercises, hand-on projects
- Few guidelines / little perspective
- Inefficient approach both for learners and teachers

RESEARCH STATEMENT

How to better understand learning processes regarding NoSQL database concepts?

Our vision (proposal paper):

1. Design a **gamified learning platform** to learn NoSQL
2. Identify **effective learning pathways** of NoSQL databases

RELATED WORK

NoSQL educational tools [T25]

- TriQL [ARZ+22]
- NoSQL e-learning laboratory [WB22]
- PrairieLearn [WHZ15]

Gamification [AF15]

- "Learning-by-doing"
- Rewards, objectives
- Traditional teaching complement

RELATED WORK

NoSQL educational tools [T25]

- TriQL [ARZ+22]
- NoSQL e-learning laboratory [WB22]
- PrairieLearn [WHZ15]

- → Lack pathway flexibility

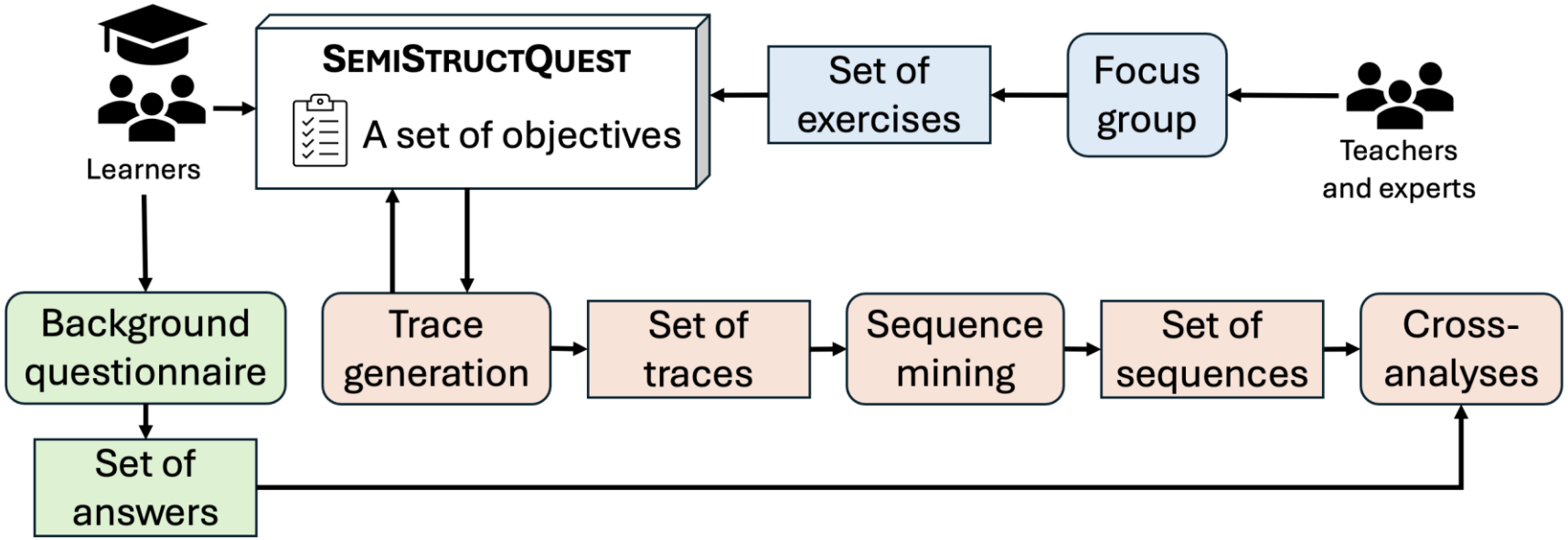
Gamification [AF15]

- "Learning-by-doing"
- Rewards, objectives
- Traditional teaching complement

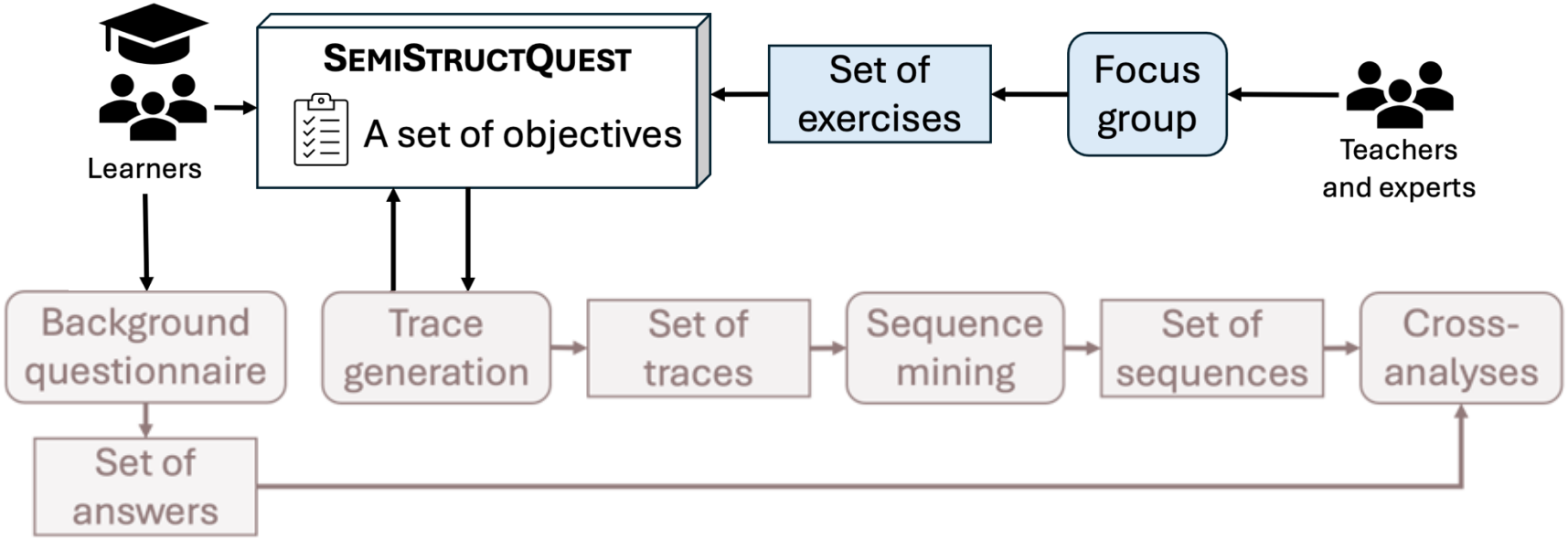
- → Lack pathway analysis

- Flexible pathway for each learner
- Data-mining based analysis of traces

OVERALL PICTURE



OVERALL PICTURE



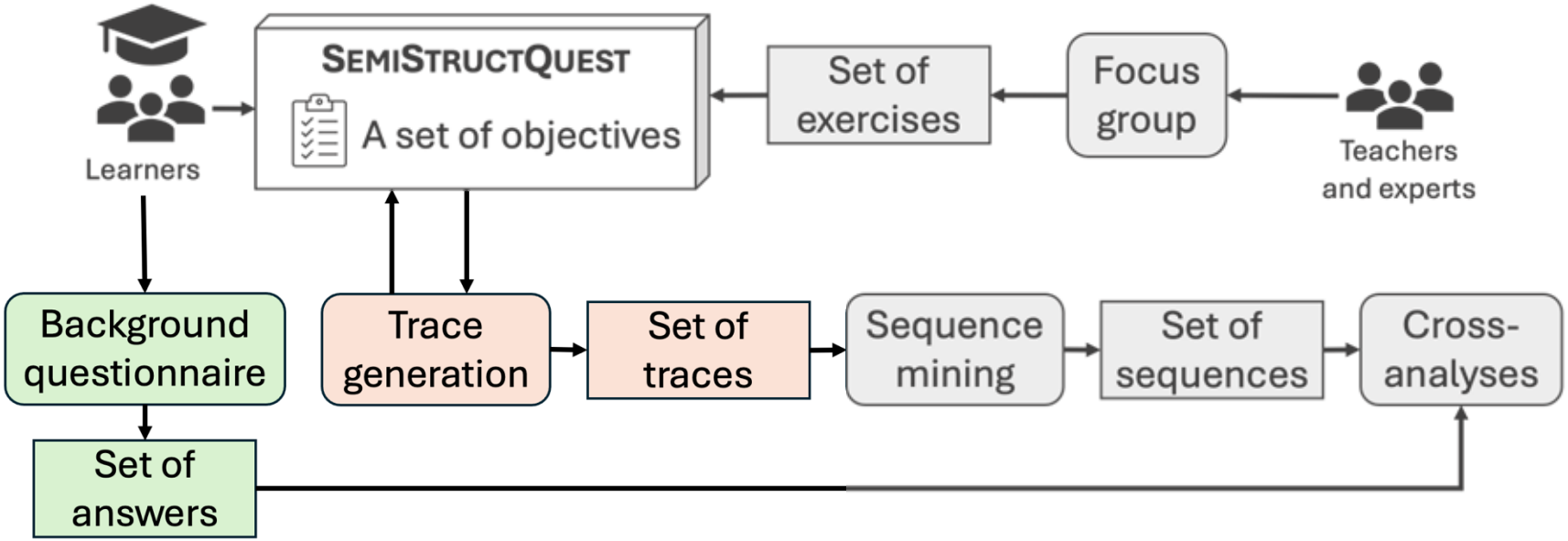
THE LEARNING PLATFORM (GOAL 1)

- **Gamified** interface
- Journey with levels and **different paths**

- Levels to be built with a 10-member **focus-group**
- **Representative exercises** with
  - Related NoSQL concepts
  - Pedagogical objectives / difficulty
  - Criteria of valid answers

Interface example

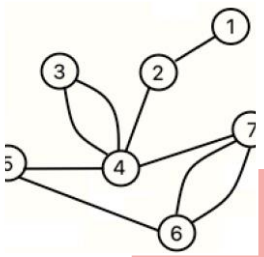
OVERALL PICTURE



PROFILES AND TRACES (GOAL 2)

PROFILES

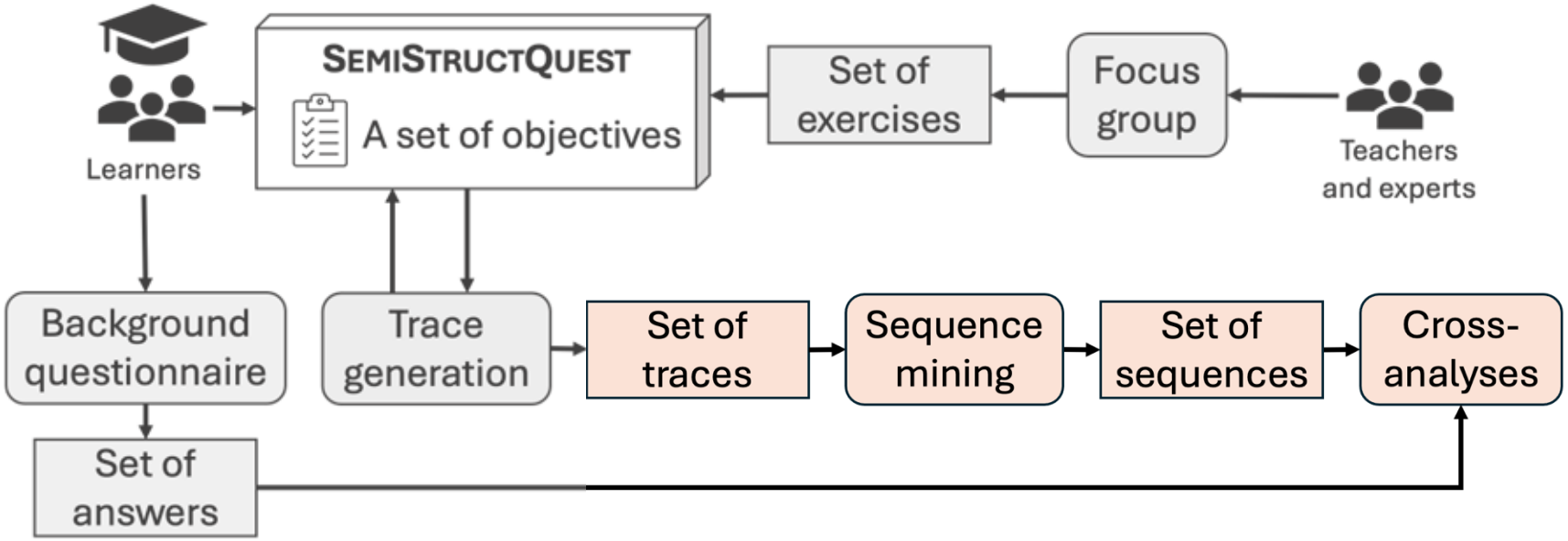
- When entering the platform
  - Any NoSQL course?
  - Any NoSQL usage?
  - Any NoSQL query?
  - Self-assessed proficiency? (Likert scale)



TRACES

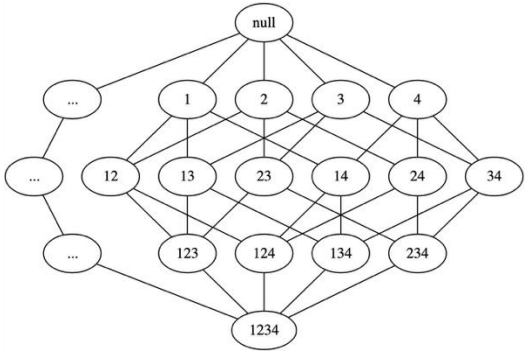
- From user interactions
- For each started exercise:
  - Duration to succeed
  - Number of tries
  - Score
- Build learning sequences for each learner

OVERALL PICTURE

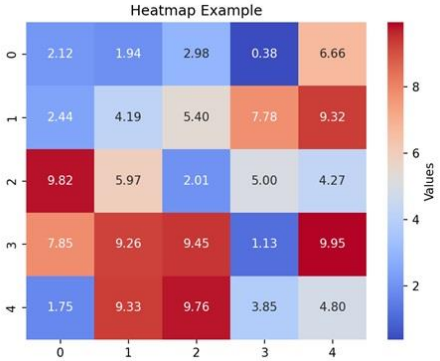


EFFECTIVE LEARNING SEQUENCES (GOAL 2)

1. Find common/uncommon learning sequences
    1. Sequential pattern mining: A-priori [AIS93]
    2. Extract most/less frequent (sub)sequences of exercises



2. Align learning sequences with learner profiles
    1. Goal: identify "optimal" learning strategies
    2. Correlate with the score, profile and learning progress
    3. Confusion matrix



CONCLUSION AND FUTURE WORK

Key takeaways:

- Gamified learning platform with flexible journey
- Automatic learning sequence extraction from traces
- Learning sequence correlation with score, profile and progress

Future work:

- Build the focus group
- Implement an initial interface
- Cluster students based on profile and traces

CONCLUSION AND FUTURE WORK

Key takeaways:

- Gamified learning platform with flexible journey
- Automatic learning sequence extraction from traces
- Learning sequence correlation with score, profile and progress

Future work:

- Build the focus group
- Implement an initial interface
- Cluster students based on profile and traces